

Notes on Algebra

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1 Gaussian Elimination

For an augmented matrix that represents a one-to-one correspondence with linear equations.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix} \longleftrightarrow \begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \cdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}, \quad (1)$$

Assuming the base field $\mathbb{F}$ so that nonzero elements are invertible, we define the transformations

Definition 1.1 (*I, II*: Elementary Row Operations).

$$I(i, k) : \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} & b_i \\ a_{k1} & a_{k2} & \cdots & a_{kn} & b_k \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix} \mapsto \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kn} & b_k \\ a_{i1} & a_{i2} & \cdots & a_{in} & b_i \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{pmatrix}.$$

$$II(i, k, c) : \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} & b_i \\ a_{k1} & a_{k2} & \cdots & a_{kn} & b_k \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix} \mapsto \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{i1} + ca_{k1} & a_{i2} + ca_{k2} & \cdots & a_{in} + ca_{kn} & b_i + cb_k \\ a_{k1} & a_{k2} & \cdots & a_{kn} & b_k \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix}.$$

Conducting operations of type *I, II*, the resulting augmented matrix represents a linear system that is equivalent to the original one.

Gaussian elimination can be standardized for an augmented matrix A ,

$$G(A) : \text{if } a_{11} = 0 \text{ and } \exists i > 1, a_{i1} \neq 0, I(1, i). \\ \forall 1 < i \leq n, II(i, 1, -a_{i1}a_{11}^{-1}).$$

We iterate $G(A)$, with the results matrix

$$\bar{A} := \begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & \cdots & \bar{a}_{1n} & b'_1 \\ 0 & \cdots & \bar{a}_{2i} & \bar{a}_{2(i+1)} & \bar{a}_{2n} & b'_2 \\ 0 & \cdots & \bar{a}_{3k} & \bar{a}_{3(k+1)} & \bar{a}_{3n} & b'_3 \\ \vdots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & \bar{a}_{rs} & \bar{a}_{r(s+1)} & \bar{a}_{rn} & b'_r \\ 0 & 0 & 0 & \cdots & 0 & b'_{r+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & b'_m \end{pmatrix}$$

in which $1 < i < k < \cdots < s$. $\bar{a}_{ij}$ denote the entries after elimination.

Theorem 1.2. *The conditions of linear equations ascertaining consistency and determinacy are as follows.*

1. *If rows such as $(0, 0, \dots, b_i \neq 0)$ do not exist in the result matrix of Gaussian elimination, then the linear equations are consistent.*
2. *If $r = n$ for consistent linear equations, they are determined.*

Proof. We mimic the regression process, observing the result matrix. The main unknown number x_{rs} is determined by other free unknown numbers $x_{r(s+1)}, x_{r(s+2)}, \dots, x_{rn}$. Similarly, $x_{(r-1)s}$ is determined by x_{rs} and $x_{r(s+1)}, x_{r(s+2)}, \dots, x_{rn}$. Hence, the system is consistent.

If $r = n$, then there are no free unknown numbers, and thus, the linear equations are determined, which means

Consistent linear equations are determined $\iff$ the result matrix in *Row Echelon Form (REF)* satisfies $r = n$.

□

From the perspective of mapping, we define

$$T_A : V \rightarrow W, \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \mapsto \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\mathbf{x} \rightarrow A\mathbf{x}}. \quad (2)$$

The result we summarized in Gaussian elimination can be transcribed as:

$$\ker T_A = \ker T_{G(A)} \quad (\text{When } b_i = 0, \text{ the } G(A) \text{ process doesn't alter the solutions of the system}) \quad (3)$$

$$\text{Im } T_A = \text{Im } T_{G(A)} \quad (\text{one-to-one correspondence can be established through the definition of } II) \quad (4)$$

Furthermore, by Gaussian elimination, we can conclude that the matrix for every T_A is row-equivalent to one *Row Echelon Form* $\bar{A}$.

$$\begin{aligned} & \forall T_A : V \mapsto W, \mathbf{x} \longrightarrow A\mathbf{x}, \\ & \iff T_{G(A)} : V \mapsto W, \mathbf{x} \longrightarrow G(A)\mathbf{x}. \end{aligned}$$

If the $r = n$, then the matrix for T_A is row-equivalent to one REF matrix such that there are no $(0, 0, \dots, b_i)$ rows.

2 Matrices & Determinants

Definition 2.1 (Determinants). *The n -order determinants are*

$$\det A = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} := \sum_{i=1}^n (-1)^{i+1} a_{i1} M_{i1}.$$

M_{ij} , the minor, is a $n - 1$ -order determinant that excludes the i^{th} row and j^{th} column of A . From the definition,

1. $\det A$ is a linear function for every row of A

2. $A \xrightarrow{I(i,k)} A', \det A = -\det A'$

Theorem 2.2.

$$A \xrightarrow{I(i,k)} A', \det A = -\det A'$$

$$A \xrightarrow{II(i,k)} A', \det A = \det A'.$$

Proof. The proof is followed by the fact that the determinants of matrix A can be expressed as a linear function of any A' 's row. It is solvable by induction. □

Theorem 2.3. From the last theorem, we discuss the effect of Gaussian elimination on determinants. For matrix $r = n$,

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \xleftrightarrow[I, II]{\text{row-equivalent}} \bar{A} = \begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & \bar{a}_{1n} \\ 0 & \bar{a}_{22} & \cdots & \bar{a}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \bar{a}_{nn} \end{pmatrix}$$

1. $\det A = \bar{a}_{11}\bar{a}_{22}\bar{a}_{33}\bar{a}_{44}\cdots\bar{a}_{nn}$.
2. The linear equations with the coefficient matrix A are consistent and determined $\iff \det A \neq 0$.

Definition 2.4 (Singular matrix). If a matrix's determinant is 0, then it is singular.

From these attributes of determinants, we are able to deduce the following.

Theorem 2.5. $\forall F : M_{m \times n}(\mathbb{R}) \rightarrow \mathbb{R}, A \mapsto F(A)$ satisfies:

1. $F(A)$ is a linear function for every row of A ,
2. $A \xrightarrow{I(i,k)} A', F(A) = -F(A')$.

$$F(A) = F(E) \det A.$$

Proof. The proof of the theorem is back substitution. We always reduce A to an upper triangular matrix. We first notice that from $A \xrightarrow{I(i,k)} A', F(A) = -F(A')$, we obtain that $F(A) = (-1)^t F(\bar{A})$ if we use I process t times since II doesn't alter $F(A)$. Thus, what's left is

$$F(\bar{A}) = F(E) \bar{a}_{11} \bar{a}_{22} \bar{a}_{33} \cdots \bar{a}_{nn},$$

in which

$$E := \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}, \det E = 1.$$

Since $F(A)$ is a linear function for every row of A ,

$$\begin{aligned} F(\bar{A}) &= \bar{a}_{nn} \cdot F \left(\begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & \bar{a}_{1n} \\ 0 & \bar{a}_{22} & \cdots & \bar{a}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \right) \\ &\xrightarrow{\forall 1 \leq i < n, II(i, n, -\bar{a}_{in})} \bar{a}_{nn} \cdot F \left(\begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & 0 \\ 0 & \bar{a}_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \right). \end{aligned}$$

The process is a replica of back substitution. Matrix $\bar{A}$ would eventually be E . □

The Column of the determinant

We consider the propagated form of the determinant's definition, which consists of eliminating a random column of the original matrix:

$$F_j(A) = \sum_{i=1}^n (-1)^{i+1} a_{ij} M_{ij}.$$

We can simply conclude that it satisfies Theorem 2.5; thus $F_j(A) = F_j(E) \det A$.

$$\begin{aligned} F_j(E) &= \sum_{i=1}^n (-1)^{i+1} a_{ij} M_{ij} = (-1)^{j+1} a_{jj} M_{jj} = (-1)^{j+1} \\ F_j(A) &= (-1)^{j+1} \det A \\ \det A &= \sum_{i=1}^n (-1)^{i+j} M_{ij} \cdot a_{ij} =: \sum_{i=1}^n A_{ij} a_{ij}. \end{aligned}$$

by induction hypothesis: $M_{jj} = \det E$ for order $n-1$, $\det E = 1$. A_{ij} is the *Algebraic cofactor* of a_{ij} . The linearity of determinants for columns, with the assistance of algebraic cofactors, implies the viability of Theorem 2.2 on columns.

Lemma 2.6 (algebraic cofactors acting on linear combinations).

$$\forall k \neq j, \sum_{i=1}^n A_{ik} a_{ij} = 0$$

Proof. We construct a matrix A' such that its k^{th} column is identical to the j^{th} column.

Now, $\det A' = \sum_{i=1}^n A'_{ik} a_{ik} = 0$, but the $A'_{ik} = A_{ik}$, $a_{ij} = a_{ik}$. □

Theorem 2.7 (Cramer's Rule). *For linear equations with the augmented matrix*

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{pmatrix}, \text{ in which } A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

the k^{th} unknown number

$$x_k = \frac{\det A_k}{\det A}.$$

The A_k is constructed by replacing the k^{th} column of A to $(b_1, b_2, \dots, b_n)$. If $\det A = 0$, it is a singular matrix and the linear equations is undetermined or inconsistent.

Proof. For x_k , we may multiply

$$\begin{pmatrix} A_{1k}a_{11} & A_{1k}a_{12} & \cdots & A_{1k}a_{1n} \\ A_{2k}a_{21} & A_{2k}a_{22} & \cdots & A_{2k}a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{nk}a_{n1} & A_{nk}a_{n2} & \cdots & A_{nk}a_{nn} \end{pmatrix}, \begin{pmatrix} A_{1k}b_1 \\ A_{2k}b_2 \\ \vdots \\ A_{nk}b_n \end{pmatrix}.$$

Then, we add every term of the linear equations. Every $\sum_{i=1}^n A_{ik}a_{ij}$ would cancel out. The only remainder term is x_k , and its coefficient is $\sum_{i=1}^n A_{ik}a_{ik} = \det A$. The right-hand side is $\sum_{i=1}^n A_{ik}b_i = \det A_k$. □

2.1 Permutations

Definition 2.8 (even(odd) permutation). *Defining the function $\varepsilon(\mathbf{J})$, $\mathbf{J} = (i_1, i_2, \dots, i_n)$ is a permutation.*

$$\varepsilon(\mathbf{J}) := \begin{cases} 1 & \text{permutation } \mathbf{J} \text{ is derived by even-numbered transposition from origin} \\ -1 & \text{permutation } \mathbf{J} \text{ is derived by odd-numbered transposition from origin} \end{cases}$$

If $\varepsilon(\mathbf{J}) = 1$, $\mathbf{J}$ is an even permutation, and vice versa.

Definition 2.9 (symmetric(antisymmetric) functions). *For a symmetric function $F(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$,*

$$F(\mathbf{a}_1, \dots, \mathbf{a}_i, \dots, \mathbf{a}_j, \dots, \mathbf{a}_n) = F(\mathbf{a}_1, \dots, \mathbf{a}_j, \dots, \mathbf{a}_i, \dots, \mathbf{a}_n).$$

For an antisymmetric function $F(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$,

$$F(\mathbf{a}_1, \dots, \mathbf{a}_i, \dots, \mathbf{a}_j, \dots, \mathbf{a}_n) = -F(\mathbf{a}_1, \dots, \mathbf{a}_j, \dots, \mathbf{a}_i, \dots, \mathbf{a}_n).$$

Theorem 2.10 (A definition of determinant). *The determinant of a n th-order matrix is a solitary function $F(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$ of n rows with n elements such that*

1. *It is a linear function for each row*
2. *It is antisymmetric*
3. *$F(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) = 1$, in which $\mathbf{e}_i$ is the row that has 1 in its i th position and 0 for its other positions.*

Theorem 2.11 (multilinear function). *Any multilinear function can be expressed as*

$$F(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n) = \sum_{(i_1, i_2, \dots, i_n)} \alpha_{i_1, i_2, \dots, i_n} a_{1i_1} a_{2i_2} a_{3i_3} \cdots a_{ni_n}$$

in which $\mathbf{a}_i = (a_{i1}, a_{i2}, a_{i3}, \dots, a_{in})$, and coefficient $\alpha_{i_1, i_2, \dots, i_n}$ are only related to their indexes rather than rows. The coefficient $\alpha_{i_1, i_2, \dots, i_n} = F(\mathbf{e}_{i_1}, \mathbf{e}_{i_2}, \dots, \mathbf{e}_{i_n})$.

(idea) It is easily proven by induction, since any linear function is $F(\mathbf{x}) = \sum_{i=1}^n a_i x_i$. In fact, from Theorem 2.11, we acknowledge the explicit formula for determinants. The determinant function

$$F(\mathbf{e}_{i_1}, \mathbf{e}_{i_2}, \dots, \mathbf{e}_{i_n})$$

can be considered a permutation $\mathbf{J}$ of the rows from the original matrix E . Since we know the I process adds $-$ to its determinant, we have $F(\mathbf{e}_{i_1}, \mathbf{e}_{i_2}, \dots, \mathbf{e}_{i_n}) = \varepsilon(\mathbf{J}) \det E = \varepsilon(\mathbf{J})$. Thus,

$$\det A = \sum_{\mathbf{J}} \varepsilon(\mathbf{J}) a_{1i_1} a_{2i_2} a_{3i_3} \cdots a_{ni_n}.$$

2.2 Rank

Definition 2.12 (Rank of a Matrix). *Let $A \in M_{n \times n}$. The rank of A , denoted by $\text{rank}(A)$, is defined as*

$$\text{rank}(A) = r$$

if and only if:

1. *There exists at least one nonzero $r \times r$ minor of A ;*
2. *Every $(r+1) \times (r+1)$ minor of A is zero.*

Equivalently, $\text{rank}(A)$ is the largest integer r for which A has a nonzero r -th order minor.

Theorem 2.13. *Let $A \in M_{m \times n}(\mathbb{F})$ and let B be obtained from A by the elementary row operation*

$$II(i, j, p).$$

Then

$$\text{rank}(A) = \text{rank}(B).$$

Proof. Write

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \xrightarrow{II(i, j, p)} B = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} + pa_{j1} & a_{i2} + pa_{j2} & \cdots & a_{in} + pa_{jn} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Assume

$$\text{rank}(A) = r.$$

Then there exists a $r \times r$ nonzero minor

$$M = \begin{vmatrix} a_{i_1 j_1} & \cdots & a_{i_1 j_r} \\ \vdots & & \vdots \\ a_{i_r j_1} & \cdots & a_{i_r j_r} \end{vmatrix} \neq 0,$$

where

$$\{i_1, \dots, i_r\} \subset \{1, \dots, m\}, \quad \{j_1, \dots, j_r\} \subset \{1, \dots, n\}.$$

Case 1: The minor does not involve row i .

Then the corresponding minor in B is unchanged, hence still nonzero. Therefore,

$$\text{rank}(B) \geq r.$$

Case 2: Every nonzero r -minor of A contains row i .

Choose such a minor M and assume $i = i_k$ for some k .

Consider the corresponding minor in B :

$$N = \begin{vmatrix} b_{i_1 j_1} & \cdots & b_{i_1 j_r} \\ \vdots & & \vdots \\ b_{i_r j_1} & \cdots & b_{i_r j_r} \end{vmatrix}.$$

Since rows

$$\mathbf{b}_i = \mathbf{a}_i + p\mathbf{a}_j,$$

and the determinant is linear in each row, we obtain

$$N = M + pM',$$

where M' is obtained from M by replacing the i -th row with the j -th row:

$$M' = \begin{vmatrix} a_{i_1 j_1} & \cdots & a_{i_1 j_r} \\ \vdots & & \vdots \\ a_{j j_1} & \cdots & a_{j j_r} \\ \vdots & & \vdots \\ a_{i_r j_1} & \cdots & a_{i_r j_r} \end{vmatrix}.$$

Now we analyze M' . If $j \in \{i_1, \dots, i_r\}$, then M' has two identical rows; hence $M' = 0$.

If $j \notin \{i_1, \dots, i_r\}$, then the row index set of M' is

$$\{i_1, \dots, i_{k-1}, j, i_{k+1}, \dots, i_r\},$$

which does not contain i .

By assumption, every nonzero r -minor must contain row i . Therefore, any r -minor not containing row i must vanish. Hence $M' = 0$.

In both cases, $N = M \neq 0$. Thus, $\text{rank}(B) \geq r$. Since the row operation is invertible (its inverse is $II(i, j, -p)$), the same argument gives $\text{rank}(A) \geq \text{rank}(B)$. Therefore,

$$\text{rank}(A) = \text{rank}(B).$$

□

From now on, we will use index-set minor notation:

$$A[I \mid J] = \begin{pmatrix} a_{i_1 j_1} & a_{i_1 j_2} & \cdots & a_{i_1 j_r} \\ a_{i_2 j_1} & a_{i_2 j_2} & \cdots & a_{i_2 j_r} \\ \vdots & \vdots & & \vdots \\ a_{i_r j_1} & a_{i_r j_2} & \cdots & a_{i_r j_r} \end{pmatrix}, \quad I = \{i_1, \dots, i_r\}, \quad J = \{j_1, \dots, j_r\}.$$

With a little observation, we noticed that the $\text{rank}(A)$ is exactly the number of non-zero rows in the resulting REF matrix of A :

$$\bar{A} = \begin{pmatrix} \bar{a}_{11} & \bar{a}_{12} & \cdots & \cdots & \bar{a}_{1n} & b'_1 \\ 0 & \cdots & \bar{a}_{2i} & \bar{a}_{2(i+1)} & \bar{a}_{2n} & b'_2 \\ 0 & \cdots & \bar{a}_{3k} & \bar{a}_{3(k+1)} & \bar{a}_{3n} & b'_3 \\ \vdots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & \bar{a}_{rs} & \bar{a}_{r(s+1)} & \bar{a}_{rn} & b'_r \\ 0 & 0 & 0 & \cdots & 0 & b'_{r+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & b'_m \end{pmatrix}, \quad \text{rank}(\bar{A}) = \text{rank}(A) = r.$$

From the equation, we can improve Theorem 1.2 to its ranked version:

Theorem 2.14 (Improve of Theorem 1.2). *Set*

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \text{Augmented Matrix } (A | B) := \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix}$$

The conditions of linear equations ascertaining consistency and determinacy are as follows.

1. If $\text{rank}(A) = \text{rank}(A | B)$, then the linear equations are consistent.
2. If $\text{rank}(A) = n$ for consistent linear equations, they are determined.

The $\text{rank}(A)$ of matrix A unfolds the amount of linear information from linear equations with coefficient A . Rank = number of independent rows.

2.3 The operations of Matrix

Definition 2.15 (Transposition).

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \Rightarrow A^T := \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}, \quad (A^T)_{ij} = a_{ji}.$$

Definition 2.16 (Multiplication).

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, B = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1l} \\ b_{21} & b_{22} & \cdots & b_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nl} \end{pmatrix}, \quad l \leq m.$$

Then,

$$(AB)_{ij} = \sum_{k=1}^m a_{ik} b_{kj} \stackrel{\text{Einstein}}{=} a_{ik} b_{kj}.$$

The multiplication of Matrices resembles Tensor Contraction, which is also conducted by taking the sum of one identical index of two multiplying tensors

$$c_{i_1 i_2 \cdots i_{n-1}} = \sum_{i_k} a_{i_1 i_2 \cdots i_k \cdots i_n} b_{i_1 i_2 \cdots i_k \cdots i_n}.$$

Theorem 2.17. *The multiplication of matrices is associative and distributive.*

$$ABC = A(BC), A(B + C) = AB + AC, (A + B)C = AC + BC.$$

$$(AB)^T = B^T A^T.$$

Lemma 2.18 (Elementary row operation in terms of Matrix Multiplication).

$$I(i, j) \iff T_{ij} := \begin{pmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots & & \vdots \\ 0 & 0 & \cdots & \mathbf{0} & \cdots & \mathbf{1} & \cdots & 0 \\ \vdots & \vdots & & \vdots & \ddots & \vdots & & \vdots \\ 0 & 0 & \cdots & \mathbf{1} & \cdots & \mathbf{0} & \cdots & 0 \\ \vdots & \vdots & & \vdots & & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{pmatrix}, \quad (T_{ij})_{rs} = \begin{cases} 1 & r = s \neq i, j \\ 1 & (r, s) = (i, j) \text{ or } (j, i) \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

$$II(i, j, c) \iff U_{ij}(c) := \begin{pmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 & \cdots & c & \cdots & 0 \\ \vdots & \vdots & & \vdots & \ddots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots & & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{pmatrix}, \quad (U_{ij}(c))_{rs} = \begin{cases} 1 & r = s \\ c & (r, s) = (i, j) \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

Put simply, we normalized the manipulations for rows to merely left multiplication of a combination of $T_{ij}, U_{ij}(c)$.

Corollary 2.19. *By left multiplying a series of $T_{ij}, U_{ij}(c)$, we are able to reduce a random matrix A to its REF. Moreover,*

$$\text{rank}(A) = \text{rank}(T_{ij}A) = \text{rank}(U_{ij}(c)A).$$

Theorem 2.20. $\forall A, B \in M_{n \times n}, \det AB = \det A \det B$.

Proof. Fixed B , we consider $F(A) = \det AB$. The sole issue is to prove its antisymmetry.

$$(AB)_{ij} = \sum a_{ik}b_{kj}. \text{ For row } i, \mathbf{c}_i = \left(\sum a_{ik}b_{k1}, \sum a_{ik}b_{k2}, \sum a_{ik}b_{k3}, \dots, \sum a_{ik}b_{kn} \right).$$

$\implies AB$ is multilinear for rows of A .

If we switch the i, j rows of A ($I(i, j)$), it is equivalent to switch i, j rows of AB .

$$\mathbf{c}_i = \left(\sum a_{jk}b_{k1}, \sum a_{jk}b_{k2}, \sum a_{jk}b_{k3}, \dots, \sum a_{jk}b_{kn} \right).$$

$\implies F(A) = -F(T_{ij}A)$. Antisymmetry is proved.

Thus, we have $F(A) = F(E) \det A$, but $F(E) = \det B$. □

Definition 2.21 (Inverse matrix). *For $A, B \in M_{n \times n}$, if $AB = E$, then $B =: A^{-1}$ is the inverse matrix of A .*

From the definition, we immediately obtain

1. $\det A \det B = \det E = 1$.
2. $\exists! A^{-1}$.
3. $\exists C \in M_{n \times n}, CA = E$. In fact, $C = A^{-1}$.

For the third property of the inverse matrix, we present $A^T(A^T)^{-1} = E$. Considering the transposition of the equation $((A^T)^{-1})^T A = E$. $C = ((A^T)^{-1})^T$ exists. Since the association law is valid, uniqueness is obvious.

Theorem 2.22 (Explicit formula of Inverse Matrix). *We use Cramer's Rule. Considering the vector multiplication of the inverse matrix $AB = E$,*

$$Ab_j = \mathbf{e}_j := (0, 0, \dots, e_j = 1, \dots, 0)^T.$$

This is a set of linear equations with the same quantity of unknown numbers, so the elements within the inverse matrix is clear:

$$b_{ij} = \frac{\det A_i}{\det A} = \frac{A_{ji}}{\det A}.$$

The A_i is constructed by replacing the i^{th} column of A with $\mathbf{e}_j$.

We summarize the Gauss elimination: a $n \times n$ non-singular matrix A can be reduced to REF by left multiplying $U_{ij}(c), T_{ij}$. In addition, we notice that

$$\begin{pmatrix} \mathbf{c}_i \\ \mathbf{c}_j \end{pmatrix} \xrightarrow{U_{ij}(1)} \begin{pmatrix} \mathbf{c}_i + \mathbf{c}_j \\ \mathbf{c}_j \end{pmatrix} \xrightarrow{U_{ji}(-1)} \begin{pmatrix} \mathbf{c}_i \\ -\mathbf{c}_i \end{pmatrix} \xrightarrow{U_{ij}(1)} \begin{pmatrix} \mathbf{c}_j \\ -\mathbf{c}_i \end{pmatrix} \xrightarrow{V_j(-1)} \begin{pmatrix} \mathbf{c}_j \\ \mathbf{c}_i \end{pmatrix} \\ \implies T_{ij} = V_j(-1)U_{ij}(1)U_{ji}(-1)U_{ij}(1),$$

where

$$III(i, \alpha) \iff V_i(\alpha) := \begin{pmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & \alpha & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & \cdots & 0 & 1 \end{pmatrix}, \quad (V)_{ii} = \alpha.$$

It multiplies α to the i th row of the matrix A .

Reducing the square matrix to REF, which is upper triangular form, we left multiply

$$\prod_{j=1}^n \prod_{i \leq j} U_{ij}(-\bar{a}_{ij} \bar{a}_{jj}^{-1}).$$

The product eliminates elements that are not on the diagonal. However, the diagonal matrix

$$\bar{A} = \begin{pmatrix} \bar{a}_{11} & 0 & \cdots & 0 \\ 0 & \bar{a}_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \bar{a}_{nn} \end{pmatrix} = \prod_{i=1}^n V_i(\bar{a}_{ii})E.$$

Thus, we have concluded that:

Theorem 2.23. $\forall A \in M_{n \times n}, \det A \neq 0, A^{-1}$ is a series of products of $V_i(\alpha), U_{ij}(c)$, which is the same as Gaussian elimination.

Corollary 2.24. Since $U_{ij}(c), V_i(\alpha)$ are I, II, III elementary row operations (They don't alter the rank of the original matrix), $\forall A \in M_{n \times n}, B \in M_{m \times m}; \det B, \det A \neq 0, \forall C \in M_{n \times m}$,

$$\text{rank}(C) = \text{rank}(ACB).$$

3 Vector Space

Definition 3.1 (Vector Space). *The vector space L is a set that satisfies the following properties:*

1. For $+$, a binary operation, the $(L, +)$ is an Abel group.
2. $\forall \text{ scalar } \alpha, \alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}$.
3. $\forall \text{ scalar } \alpha, \beta, (\alpha + \beta)\mathbf{x} = \alpha\mathbf{x} + \beta\mathbf{x}$.

4. $\alpha\beta\mathbf{x} = \alpha(\beta\mathbf{x})$
5. $\forall \mathbf{x} \in L, 1\mathbf{x} = \mathbf{x}$

We therefore present some supplements to the vector space.

Definition 3.2 (Definitions Involving Subspace).

1. $\forall L' \subset L, s.t. \forall \mathbf{x}, \mathbf{y} \in L', \alpha\mathbf{x} + \beta\mathbf{y} \in L'. L'$ is the subspace of L .
2. L is the sum of its subspaces $L_1, L_2, \dots, L_k \iff \forall \mathbf{x} \in L, \exists \mathbf{x}_1 \in L_1, \mathbf{x}_2 \in L_2, \dots, \mathbf{x}_k \in L_k,$
 $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2 + \dots + \mathbf{x}_k.$
 $L = L_1 + L_2 + \dots + L_k.$
3. L is the direct sum of its subspaces $L_1, L_2, \dots, L_k$
 $\iff L$ is the sum of subspaces $L_1, L_2, \dots, L_k. \mathbf{x}'$'s representation in $\mathbf{x}_1 \in L_1, \mathbf{x}_2 \in L_2, \dots, \mathbf{x}_k \in L_k$ is unique.
 $L = L_1 \oplus L_2 \oplus \dots \oplus L_k.$

Definition 3.3 (Linear Independence/dependence). For vectors $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_n$, if $\exists \alpha_1, \alpha_2, \dots, \alpha_n$ such that

$$\alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2 + \alpha_3\mathbf{x}_3 + \dots + \alpha_n\mathbf{x}_n = 0$$

for at least one $\alpha_i \neq 0$. $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_n$, are linear dependent.

On the other hand, if

$$\alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2 + \alpha_3\mathbf{x}_3 + \dots + \alpha_n\mathbf{x}_n = 0 \iff \alpha_1, \alpha_2, \dots, \alpha_n = 0,$$

$\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_n$ are linear independent.

Definition 3.4 (Dimension of vector space). Similar to the $\text{rank}(A)$ in matrices, the dim is the maximum quantity of linearly independent vectors in vector space L . If $d = \text{dim } L$, then $\forall \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_{d+1}, \exists \alpha_1, \alpha_2, \dots, \alpha_{d+1}$ such that

$$\begin{aligned} \alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2 + \alpha_3\mathbf{x}_3 + \dots + \alpha_{d+1}\mathbf{x}_{d+1} &= 0. \\ \alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2 + \dots + \alpha_{i-1}\mathbf{x}_{i-1} + \alpha_{i+1}\mathbf{x}_{i+1} + \dots + \alpha_{d+1}\mathbf{x}_{d+1} &= \mathbf{x}_i \end{aligned}$$

for at least one $\alpha_i \neq 0$. There is one $\mathbf{x}_i$ that is composed by other vectors.

Lemma 3.5. As in corollary 2.24,

$$\dim(\{A\mathbf{x} : \mathbf{x} \in V\}) = \dim(\{U_{ij}(c)AU_{i'j'}^T(c')\mathbf{x} : \mathbf{x} \in V\}).$$

3.1 Vector bases and Vector Space Operations

Definition 3.6 (bases). $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n \in L$, such that they are linear independent, and $\forall \mathbf{x} \in L, \mathbf{x} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \dots + a_n\mathbf{e}_n$.

We are able to immediately derive the following fact that

Theorem 3.7. For any group of linear independent vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_m$, we can find other $\mathbf{e}_m, \mathbf{e}_{m+1}, \dots, \mathbf{e}_n$ such that they form a bases of L .

The result is proved by simply adding linear independent vectors repeatedly, and it can be written in terms of vector spaces $\forall L' \subset L, \exists L'' \subset L, L = L' \oplus L''$. The next lemma indicates the dimension's restriction on the vector space.

Lemma 3.8. Any linear combinations $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_m$ of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ such that $m > n$ are linear dependent.

Proof. Since the $\mathbf{y}_i$ is a linear combination of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$, we have

$$\begin{cases} a_{11}\mathbf{x}_1 + a_{12}\mathbf{x}_2 + \dots + a_{1n}\mathbf{x}_n = \mathbf{y}_1 \\ a_{21}\mathbf{x}_1 + a_{22}\mathbf{x}_2 + \dots + a_{2n}\mathbf{x}_n = \mathbf{y}_2 \\ \dots \\ a_{m1}\mathbf{x}_1 + a_{m2}\mathbf{x}_2 + \dots + a_{mn}\mathbf{x}_n = \mathbf{y}_m \end{cases}.$$

We have to construct m coefficients $\alpha_1, \alpha_2, \dots, \alpha_m$ that are not all zero and

$$\alpha_1\mathbf{y}_1 + \alpha_2\mathbf{y}_2 + \dots + \alpha_m\mathbf{y}_m = 0. \quad (7)$$

If we plug (8) into the linear combinations, we observe

$$(a_{11}\alpha_1 + a_{21}\alpha_2 + \dots + a_{m1}\alpha_m)\mathbf{x}_1 + (a_{12}\alpha_1 + a_{22}\alpha_2 + \dots + a_{m2}\alpha_m)\mathbf{x}_2 + \dots + (a_{1n}\alpha_1 + a_{2n}\alpha_2 + \dots + a_{mn}\alpha_m)\mathbf{x}_n = 0.$$

The result yields that as ensure the validity of the equation (it equals 0), thus we must have

$$\begin{cases} a_{11}\alpha_1 + a_{21}\alpha_2 + \dots + a_{m1}\alpha_m = 0 \\ a_{12}\alpha_1 + a_{22}\alpha_2 + \dots + a_{m2}\alpha_m = 0 \\ \dots \\ a_{1n}\alpha_1 + a_{2n}\alpha_2 + \dots + a_{mn}\alpha_m = 0 \end{cases}.$$

We solve as by Theorem 1.2 since there are fewer equations (m) than unknown numbers (n). □

Corollary 3.9. *If $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n \in L$ are bases, then $\dim L = n$.*

Next, we follow the train of thought that the bases is correlated with dimensions.

Theorem 3.10. *L_1, L_2 is a vector space.*

$$\dim(L_1 \oplus L_2) = \dim L_1 + \dim L_2,$$

$$\dim(L_1 + L_2) = \dim L_1 + \dim L_2 - \dim(L_1 \cap L_2).$$

Proof. Let $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ and $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m$ denote a bases of L_1, L_2 . Then, every vector in $L_1 \oplus L_2$ is

$$\mathbf{x}_1 + \mathbf{x}_2 = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \dots + a_n\mathbf{e}_n + a_{n+1}\mathbf{f}_1 + a_{n+2}\mathbf{f}_2 + \dots + a_{n+m}\mathbf{f}_m,$$

$\mathbf{x}_1 \in L_1, \mathbf{x}_2 \in L_2$. $\mathbf{x}_1 + \mathbf{x}_2 = 0$ yields the $a_1, a_2, \dots, a_{n+m} = 0$ since $\mathbf{x}_1, \mathbf{x}_2 = 0$, which indicates the linear independence of vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ and $\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m$.

The second result can be derived if we set $L_1 \cap L_2 = L_3, L_1 = L'_1 \oplus L_3, L_2 = L'_2 \oplus L_3$. □

Definition 3.11. *Let V be a finite-dimensional vector space over a field $\mathbb{F}$.*

A flag in V is a sequence of subspaces

$$\mathbf{0} = V_0 \subset V_1 \subset V_2 \subset \dots \subset V_k \subset V$$

such that each inclusion is strict.

If, in addition, $\dim V_i = i$ for all $i = 0, 1, \dots, n$, where $n = \dim V$ and

$$\mathbf{0} = V_0 \subset V_1 \subset \dots \subset V_n = V,$$

then the flag is called a complete flag.

The series of linear spaces is an additive process of new linear independent vectors.

3.2 Operators

Definition 3.12 (Operator). *The linear transformation, or operator, is a map $A : L \rightarrow M$ such that*

$$A(\mathbf{x} + \mathbf{y}) = A(\mathbf{x}) + A(\mathbf{y}), \quad A(\alpha\mathbf{x}) = \alpha A(\mathbf{x}).$$

The operator is an embodiment of linear algebra in which it not only fully inherits the attributes of matrix algebra, but also surpasses the axis-dependence and is dedicated to the discussion of the algebraic properties of linear spaces.

The operator corresponds to a transformation by a matrix, while a base from L, M is selected, respectively. Let $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ and $\mathcal{F} = (\mathbf{f}_1, \dots, \mathbf{f}_m)$ be bases of L, M , respectively. For any vector in L

$$\begin{aligned} \mathbf{x} &= \alpha_1 \mathbf{e}_1 + \alpha_2 \mathbf{e}_2 + \dots + \alpha_n \mathbf{e}_n, \\ A(\mathbf{x}) &= \alpha_1 A(\mathbf{e}_1) + \alpha_2 A(\mathbf{e}_2) + \dots + \alpha_n A(\mathbf{e}_n). \end{aligned}$$

If the operator transforms the bases of L into

$$\begin{cases} A(\mathbf{e}_1) = a_{11}\mathbf{f}_1 + a_{21}\mathbf{f}_2 + \dots + a_{m1}\mathbf{f}_m \\ A(\mathbf{e}_2) = a_{12}\mathbf{f}_1 + a_{22}\mathbf{f}_2 + \dots + a_{m2}\mathbf{f}_m \\ \dots \\ A(\mathbf{e}_n) = a_{1n}\mathbf{f}_1 + a_{2n}\mathbf{f}_2 + \dots + a_{mn}\mathbf{f}_m \end{cases}, \quad [A]_{\mathcal{F} \leftarrow \mathcal{E}} := \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

then we have if $A(\mathbf{x}) = \beta_1 \mathbf{f}_1 + \beta_2 \mathbf{f}_2 + \dots + \beta_m \mathbf{f}_m$,

$$\begin{cases} \beta_1 = a_{11}\alpha_1 + a_{12}\alpha_2 + \dots + a_{1n}\alpha_n \\ \beta_2 = a_{21}\alpha_1 + a_{22}\alpha_2 + \dots + a_{2n}\alpha_n \\ \dots \\ \beta_m = a_{m1}\alpha_1 + a_{m2}\alpha_2 + \dots + a_{mn}\alpha_n \end{cases} \implies [A(\mathbf{x})]_{\mathcal{F}} = [A]_{\mathcal{F} \leftarrow \mathcal{E}} [\mathbf{x}]_{\mathcal{E}} \quad (8)$$

$[A]_{\mathcal{F} \leftarrow \mathcal{E}}$ is solely determined by the bases $\mathcal{E}, \mathcal{F}$, and the operator itself. By converting to matrix $[A]_{\mathbf{e} \rightarrow \mathbf{f}}$, we obtain the trivial but important result. Here, we have just introduced a crucial notation $[\mathbf{x}]_{\mathcal{E}}$; it represents the "coordinate" of an unaltered vector $\mathbf{x}$ under a specific bases $\mathcal{E}$. For example, the scalars $\beta_1, \beta_2, \dots, \beta_m$ in (9) is the coordinates of $A(\mathbf{x})$ under $\mathcal{F}$.

Lemma 3.13. $A : L \rightarrow N, B : N \rightarrow M$. $(BA)(\mathbf{x}) := B(A(\mathbf{x}))$. Let $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$, $\mathcal{G} = (\mathbf{g}_1, \dots, \mathbf{g}_l)$ and $\mathcal{F} = (\mathbf{f}_1, \dots, \mathbf{f}_m)$ be bases of L, N, M , respectively. We reached that

$$(BA)(\mathbf{e}_i) = \sum_{j=1}^m c_{ij} \mathbf{f}_j, \quad c_{ij} = ([B]_{\mathcal{F} \leftarrow \mathcal{G}} [A]_{\mathcal{G} \leftarrow \mathcal{E}})_{ij}.$$

In addition, the $A(B + C) = AB + AC, (A + B)C = AC + BC$.

From the fundamental discussion of operators, we instantly resolve the problem involving any shift in coordinates(bases; they are actually equivalent).

Theorem 3.14 (General Shifts in bases). *Let $\mathcal{E} = (\mathbf{e}_1, \dots, \mathbf{e}_n)$, and $\mathcal{E}' = (\mathbf{e}'_1, \dots, \mathbf{e}'_n)$ be two bases of L . Since the shift in bases does not change the value of a vector, we regard the shift in bases as an Identity Operator $I_L : L \rightarrow L$. Then, the matrices*

$$[I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}^{-1} = [I_L]_{\mathcal{E} \leftarrow \mathcal{E}'}$$

Proof. We recall the expression in (9). Let $\mathbf{x} \in L$. If tuple $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$ transforms to $\mathcal{E}' = (\mathbf{e}'_1, \mathbf{e}'_2, \dots, \mathbf{e}'_n)$, then the coordinates alter as

$$[\mathbf{x}]_{\mathcal{E}'} = [I_L(\mathbf{x})]_{\mathcal{E}'} = [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}} [\mathbf{x}]_{\mathcal{E}},$$

which implies that $[I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}^{-1} = [I_L]_{\mathcal{E} \leftarrow \mathcal{E}'}$. □

Theorem 3.15 (Operators' matrix with bases Shifts). $A : L \rightarrow M$ is an operator. Let $\mathcal{E} = (e_1, \dots, e_n)$, and $\mathcal{E}' = (e'_1, \dots, e'_n)$ be two bases of L ; $\mathcal{F} = (f_1, \dots, f_n)$, and $\mathcal{F}' = (f'_1, \dots, f'_n)$ be two bases of M .

$$\begin{array}{ccc} [x]_{\mathcal{E}'} & \xleftarrow{I_L: L \rightarrow L} & [x]_{\mathcal{E}} \\ \downarrow A: L \rightarrow M & & \downarrow A: L \rightarrow M \\ [A(x)]_{\mathcal{F}'} & \xleftarrow{I_M: M \rightarrow M} & [A(x)]_{\mathcal{F}} \end{array}$$

From the diagram, we acquire the relation of matrices

$$[A]_{\mathcal{F}' \leftarrow \mathcal{E}'} = [I_M]_{\mathcal{F}' \leftarrow \mathcal{F}} [A]_{\mathcal{F} \leftarrow \mathcal{E}} [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}^{-1}.$$

Proof. The main idea is shown in the commutative diagram. In fact, by Lemma 1.39 and Theorem 1.41, we obtain

$$\begin{aligned} \begin{cases} [A(x)]_{\mathcal{F}'} = [A]_{\mathcal{F}' \leftarrow \mathcal{E}'} [x]_{\mathcal{E}'} = [A]_{\mathcal{F}' \leftarrow \mathcal{E}'} [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}} [x]_{\mathcal{E}} \\ [A(x)]_{\mathcal{F}} = [A]_{\mathcal{F} \leftarrow \mathcal{E}} [x]_{\mathcal{E}} \\ [A(x)]_{\mathcal{F}'} = [I_M]_{\mathcal{F}' \leftarrow \mathcal{F}} [A(x)]_{\mathcal{F}} \end{cases} & \implies [A]_{\mathcal{F}' \leftarrow \mathcal{E}'} [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}} [x]_{\mathcal{E}} = [I_M]_{\mathcal{F}' \leftarrow \mathcal{F}} [A]_{\mathcal{F} \leftarrow \mathcal{E}} [x]_{\mathcal{E}} \\ & \implies [A]_{\mathcal{F}' \leftarrow \mathcal{E}'} [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}} = [I_M]_{\mathcal{F}' \leftarrow \mathcal{F}} [A]_{\mathcal{F} \leftarrow \mathcal{E}} \end{aligned}$$

Because I is mapping $L \rightarrow L$, the matrix $[I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}$ must be a square matrix with a non-zero determinant. Thus, any $[I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}$ is nonsingular. $\square$

Corollary 3.16. We note that if A maps $L \rightarrow L$, then the result is reduced to

$$[A]_{\mathcal{E}' \leftarrow \mathcal{E}'} = [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}} [A]_{\mathcal{E} \leftarrow \mathcal{E}} [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}^{-1}.$$

The result means that for matrices M', M, P such that $M' = PMP^{-1}$, M, M' belong to one operator.

Remark 3.17. The book usually employs the other perspective of bases shifts, which is, without introducing I_L , constructing a "shift" operator $P : L \rightarrow L$ that directly turns the bases. $P(e_i) = e'_i$. Then, $[P(x)]_{\mathcal{E}'} = [x]_{\mathcal{E}}$.

Definition 3.18 (Determinant of an Operator). For any operator $A : L \rightarrow L$, its determinant is defined as $\det A := \det[A]_{\mathcal{E} \leftarrow \mathcal{E}}$ for any bases.

3.3 Linear Space Algebra

Definition 3.19 (Isomorphism). If operator $A : L \rightarrow M$ is also a bijection (One-to-one correspondence), then it is an isomorphism. We denote $A : L \xrightarrow{\sim} M$, and $L \cong M$. The Isomorphism is an equivalence relation.

Theorem 3.20. If linear spaces L, M have limited dimensions, $L \cong M \iff \dim L = \dim M$.

Proof. The necessary condition of the theorem, $\dim L = \dim M \Rightarrow L \cong M$, can be proved by using the $\mathbb{K}^n := \{(a_1, a_2, \dots, a_n) : a_1, a_2, \dots, a_n \in \mathbb{K}\}$. It clear that $L \cong \mathbb{K}^n, M \cong \mathbb{K}^n$, and $L \cong M$.

The sufficient condition requires that for a bases $e_1, e_2, \dots, e_n \in L$, $f_1, f_2, \dots, f_n = A(e_n) \in M$ is also a bases when $L \cong M$, which in turn indicates that $f_1, f_2, \dots, f_n$ are linear independent, and $\forall x \in M$, x is a linear combination of $f_1, f_2, \dots, f_n$. Let $A \xrightarrow{\sim} M$.

If $a_1 e_1 + a_2 e_2 + \dots + a_n e_n = \mathbf{0}$, then $A(a_1 e_1 + a_2 e_2 + \dots + a_n e_n) = A(\mathbf{0})$, $a_1 f_1 + a_2 f_2 + \dots + a_n f_n = \mathbf{0}' \in M$, but such a scenario only takes place while $a_1, a_2, \dots, a_n = 0$ since $A(\mathbf{0}) = \mathbf{0}'$. Thus, $f_1, f_2, \dots, f_n$ are linear independent.

For any $x \in M$, $\exists! y \in L, A(y) = x$. $y = a_1 e_1 + a_2 e_2 + \dots + a_n e_n$, so $x = A(a_1 e_1 + a_2 e_2 + \dots + a_n e_n) = a_1 f_1 + a_2 f_2 + \dots + a_n f_n$. Thus, $\forall x \in M$, x is a linear combination of $f_1, f_2, \dots, f_n$. $\square$

Definition 3.21 (Kernel). For $A : L \rightarrow M$, $\ker A := \{x \in L : A(x) = \mathbf{0}' \in M\}$. $\ker A \subset L$.

Definition 3.22 (Image). For $A : L \rightarrow M$, $\text{Im } A := \{A(x) : x \in L\}$. $\text{Im } A \subset M$.

It is time to introduce the Rank-Nullity theorem to reveal the correlation between our previous insights, which we have already seen in the Gaussian elimination and Rank chapters.

Theorem 3.23 (Rank-Nullity Theorem). *For any linear transformation such that $T : V \mapsto W$, $\dim V$ is finite,*

$$\dim(\text{Im } T) + \dim(\ker T) = \dim V \quad (9)$$

The creation of the theorem is inspired by (3), (4). We can easily verify some interesting facts: for $[T(\mathbf{x})]_{\mathcal{E}'} = A[\mathbf{x}]_{\mathcal{E}}$, $\text{rank}(A) = \dim(\text{Im } T)$; $\dim(\ker T)$ is defined as the *nullity* of the mapping T .

We are able to provide a draft addressing the validity of the Rank-Nullity Theorem, *in my words*.

Proof. The linear map T is depicted by a matrix A that transforms the vectors in V to W ,

$$T : V \rightarrow W, [\mathbf{x}]_{\mathcal{E}} \mapsto [T]_{\mathcal{E}' \leftarrow \mathcal{E}} [\mathbf{x}]_{\mathcal{E}},$$

$$A := [T]_{\mathcal{E}' \leftarrow \mathcal{E}}, \mathbf{b} := A[\mathbf{x}]_{\mathcal{E}}.$$

If we choose another bases $\mathcal{F}'$, $\mathcal{E}'$, then according to Theorem 1.40:

$$A' = [I_M]_{\mathcal{F}' \leftarrow \mathcal{F}} A [I_L]_{\mathcal{E}' \leftarrow \mathcal{E}}^{-1} \implies \text{rank}(A') = \text{rank}(A), \text{ which implies the inalterableness of rank.}$$

$$\mathbf{x} := (x_1, x_2, \dots, x_n)^T \in V, \mathbf{b} := (b_1, b_2, \dots, b_m)^T \in \text{Im } T \subset W, A \in M^{m \times n}.$$

We shall provide some insight for the proof of the theorem one step at a time.

1. $\dim V = n$, n represents the dimension of the domain V , which is the number of coordinates of the input vector $\mathbf{x}$.

2. $\dim(\text{Im } T) = \text{rank}(A)$.

The $\dim(\text{Im } T)$ represents the number of vectors in a bases in $\text{Im } T = \{A\mathbf{x} : \mathbf{x} \in V\}$. We incorporate lemma 1.30 to reduce the matrix A to Row Echelon Form, and we notice that the result vector of $\mathbf{b}$ is solely determined by the $\text{rank}(A)$ main unknown numbers. Thus, $\dim(\text{Im } T) = \text{rank}(A)$.

3. $\dim(\ker T) =$ the quantity of free unknown numbers in the linear equations $= n - \text{rank}(A)$.

The $\ker T \subset V$ is a linear subspace. Observing the Row Echelon Form of A , we realize that for each $\mathbf{x}$ such that $T(\mathbf{x}) = \mathbf{0}$, its pivot unknown numbers (the first $\text{rank}(A)$ unknown numbers) are solely determined by its last $n - \text{rank}(A)$ free unknown numbers according to the linear equations. Thus, its dimension is exactly $n - \text{rank}(A)$.

Therefore, we obtain $\dim(\text{Im } T) + \dim(\ker T) = \text{rank}(A) + n - \text{rank}(A) = n = \dim V$. □

Remark 3.24. *The rank-nullity theorem is actually a linear-algebraic version of Isomorphism Theorem A that puts dimension on both sides.*

$$G / \ker \sigma \cong \text{Im } \sigma \implies V / \ker T \cong \text{Im } T, \quad (10)$$

for which

$$\dim V / \ker T = \dim V - \dim(\ker T).$$

My proof is based on rather specific calculations on Matrix; however, the theorem can also be proven in pure operator algebra, refining with extra universality. For such proof to be done, we have to develop our knowledge toward isomorphism.

Theorem 3.25 (Isomorphic Condition involving Kernel).

$$\forall A : L \rightarrow M, \dim L = \dim M, \ker A = \{\mathbf{0}\} \implies A : L \xrightarrow{\sim} M.$$

Just consider the bases $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n \in L$, and $A(\mathbf{e}_1), A(\mathbf{e}_2), \dots, A(\mathbf{e}_n) \in M$.

Theorem 3.26 (Isomorphic Condition involving Image).

$$\forall A : L \rightarrow M, \dim L = \dim M, \text{Im } A = M \implies A : L \xrightarrow{\sim} M.$$

Consider bases $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n \in L$ such that $A(\mathbf{e}_1), A(\mathbf{e}_2), \dots, A(\mathbf{e}_n) \in M$ forms a bases.

Theorem 3.27 (Isomorphic Condition involving Singularity). *If for every bases $\mathcal{E} \in L$, $\mathcal{F} \in M$, $A : L \rightarrow M$ has $\det([A]_{\mathcal{F} \leftarrow \mathcal{E}}) \neq 0$, then A is nonsingular.*

$$A : L \xrightarrow{\sim} M, \dim L = \dim M \iff A \text{ is nonsingular.}$$

Definition 3.28 (Rank of an operator). For $A : L \rightarrow M$, $\text{rank}(A) := \dim(\text{Im } A)$.

Now, we have enough prerequisites to provide a general proof of the Rank-Nullity theorem.

Theorem 3.29 (Extended Rank-Nullity Theorem). For an operator $A : L \rightarrow M$, $\text{rank}(A) + \dim(\ker A) = \dim L$. Plus, we are able to select bases $\mathcal{E}, \mathcal{F}$ in L, M such that

$$[A]_{\mathcal{F} \leftarrow \mathcal{E}} = \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix}, \quad r := \text{rank}(A).$$

Proof. We decompose $L = \ker A \oplus L_1$, $M = \text{Im } A \oplus M_1$. Considering the restriction of A on $L_1 \rightarrow \text{Im } A$, namely $A' : L_1 \rightarrow \text{Im } A$, $\forall \mathbf{x} \in L, \mathbf{x} = \mathbf{u} + \mathbf{v}$, $\mathbf{u} \in \ker A, \mathbf{v} \in L_1$, $A(\mathbf{x}) = A(\mathbf{u}) + A(\mathbf{v}) = \mathbf{0} + A'(\mathbf{v})$. Noticing that $\ker A' = \{\mathbf{0}\}$, $\text{Im } A' = \text{Im } A$,

$$\forall \text{ base } \mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_r \in \text{Im } A, \exists \mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_r \in L_1, \mathbf{f}_i = A'(\mathbf{e}_i), \text{ is also a base } \implies A' : L_1 \xrightarrow{\sim} \text{Im } A.$$

We derived $L_1 \cong \text{Im } A$, and $\dim L = \dim(\ker A) + \dim L_1 = \dim(\ker A) + \dim(\text{Im } A)$.

We select base $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in L_1$, and $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{n-r} \in \ker A$. Combining, we obtain $\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{u}_1, \dots, \mathbf{u}_{n-r}$ as another base of L .

Since A' is an isomorphism, $A'(\mathbf{v}_1), A'(\mathbf{v}_2), \dots, A'(\mathbf{v}_r) \in \text{Im } A$, and we just expand these linear independent vectors to a base in M . Thus, for a base $\mathcal{F} = A'(\mathbf{v}_1), A'(\mathbf{v}_2), \dots, A'(\mathbf{v}_r), \mathbf{f}_{r+1}, \dots, \mathbf{f}_m \in M$, we found a base $\mathcal{E} = \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{n-r} \in L$ such that it satisfied

$$[A]_{\mathcal{F} \leftarrow \mathcal{E}} = \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix}.$$

□

Corollary 3.30. $\forall A = M^{m \times n}, \exists \text{ nonsingular } B, C \text{ such that}$

$$BAC^{-1} = \begin{pmatrix} E_r & 0 \\ 0 & 0 \end{pmatrix}$$

3.4 Dual space

4 A Short summary from Gaussian elimination to vector space

Theorem 4.1 (1605, Binet-Cauchy Formula). $A \in M^{m \times n}, B \in M^{n \times m}$.

$$C = AB = (c_{ij}), \quad c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}, \quad (i, j = 1, \dots, m),$$

$A_{i_1, \dots, i_m}$ selects i columns of matrix A ; $B_{i_1, \dots, i_m}$ extracts B 's columns.

$$\det(C) = \sum_{1 \leq i_1 < \dots < i_m \leq n} \det A_{i_1, \dots, i_m} \det B_{i_1, \dots, i_m}.$$

If $m > n$, $\det C = 0$.

Proof. Denote by

$$\mathcal{I} = \{(i_1, \dots, i_m) : 1 \leq i_1 < \dots < i_m \leq n\}.$$

1. Leibniz expansion of $\det C$.

Using Leibniz formula,

$$\det C = \sum_{\sigma \in S_m} \varepsilon(\sigma) \prod_{l=1}^m c_{l\sigma(l)} = \sum_{\sigma \in S_m} \varepsilon(\sigma) \prod_{l=1}^m \sum_{k=1}^n a_{lk} b_{k\sigma(l)}.$$

$$\det C = \sum_{\sigma \in S_m} \varepsilon(\sigma) \sum_{1 \leq k_1, \dots, k_m \leq n} \prod_{l=1}^m a_{lk_l} b_{k_l \sigma(l)},$$

by expanding the product.

2. Cancellation of terms with repeated indices.

If $(k_1, \dots, k_m)$ contains repeated values, say $k_p = k_q$ with $p \neq q$, then exchanging p and q does not change the product

$$\prod_{l=1}^m a_{lk_l} b_{k_l \sigma(l)},$$

but changes the sign of the permutation σ . Hence such terms cancel in pairs.

Therefore, only tuples $(k_1, \dots, k_m)$ with all k_l distinct contribute.

3. Reindexing by subsets and permutations.

If $k_1, \dots, k_m$ are all distinct, then they form a subset $I = \{i_1, \dots, i_m\} \in \mathcal{I}$, and there exists a unique permutation $\tau \in S_m$ such that $k_l = i_{\tau(l)}$. Thus, we may rewrite

$$\begin{aligned} \sum_{k_1, \dots, k_m \text{ distinct}} &\iff \sum_{I \in \mathcal{I}} \sum_{\tau \in S_m} \\ \det C &= \sum_{\sigma \in S_m} \varepsilon(\sigma) \sum_{I \in \mathcal{I}} \sum_{\tau \in S_m} \prod_{l=1}^m a_{li_{\tau(l)}} b_{i_{\tau(l)} \sigma(l)}. \end{aligned}$$

4. Rearranging permutations.

Let $\rho := \sigma \circ \tau^{-1}$. Then $\sigma = \rho \circ \tau$, and $\varepsilon(\sigma) = \varepsilon(\rho)\varepsilon(\tau)$. Hence

$$\det C = \sum_{I \in \mathcal{I}} \sum_{\rho \in S_m} \varepsilon(\rho) \left(\sum_{\tau \in S_m} \varepsilon(\tau) \prod_{l=1}^m a_{li_{\tau(l)}} \right) \left(\prod_{k=1}^m b_{i_k \rho(k)} \right).$$

5. Identification with determinants.

Observe that

$$\begin{aligned} \sum_{\tau \in S_m} \varepsilon(\tau) \prod_{l=1}^m a_{li_{\tau(l)}} &= \det(A_{i_1, \dots, i_m}), \\ \sum_{\rho \in S_m} \varepsilon(\rho) \prod_{k=1}^m b_{i_k \rho(k)} &= \det(B_{i_1, \dots, i_m}). \end{aligned}$$

Therefore,

$$\det C = \sum_{I \in \mathcal{I}} \det(A_{i_1, \dots, i_m}) \det(B_{i_1, \dots, i_m}).$$

If $m > n$, there are no such subsets I , hence $\det C = 0$. □